

[5/A-3]

SARDAR PATEL UNIVERSITY**BCA Sem-II EXAMINATION (NC)****Date: 07/09/2020****Time: 10.00 am to 12.00 noon****SUBJECT: MATHEMATICS-II (US02FBCA02)****Maximum Marks: 70**

Q:1 Choose the correct option in the following, mention the correct option with the answers in the answer book. [10]

- (1) The number of edges in the complete graph K_{10} is
 - (a) 55
 - (b) 20
 - (c) 45
 - (d) 10
- (2) An alternating sequence of vertices and edges in graph is called
 - (a) trail
 - (b) cycle
 - (c) path
 - (d) degree
- (3) The degree of pendant vertex is
 - (a) one
 - (b) two
 - (c) zero
 - (d) three
- (4) In a connected map with $E = 6$, $R = 3$ then $V =$ ____
 - (a) 2
 - (b) 5
 - (c) 4
 - (d) 6
- (5) $\binom{n}{n-7} =$
 - (a) $\binom{n}{n+7}$
 - (b) $\binom{n}{n}$
 - (c) $\binom{n}{7}$
 - (d) $\binom{n}{0}$
- (6) Which of the following is the superior measure of Dispersion
 - (a) Range
 - (b) Standard deviation
 - (c) Mean deviation
 - (d) Quartile deviation
- (7) Four persons out of five persons can be arranged in a row in ways
 - (a) 5
 - (b) 10
 - (c) 120
 - (d) 24
- (8) A square of is called Variance
 - (a) Range
 - (b) Mean deviation
 - (c) Standard deviation
 - (d) Quartile deviation
- (9) The relation between correlation coefficient and regression coefficients is
 - (a) $r = \pm \sqrt{b_{XY} + b_{YX}}$
 - (b) $r = \frac{b_{XY} + b_{YX}}{2}$
 - (c) $r = \pm \sqrt{b_{XY} b_{YX}}$
 - (d) None of these
- (10) A graph that can be drawn in a plane or on a sphere so that its edge do not cross is said to be ____
 - (a) planar
 - (b) non planar
 - (c) complete
 - (d) simple

Q:2 Do as Directed:

[08]

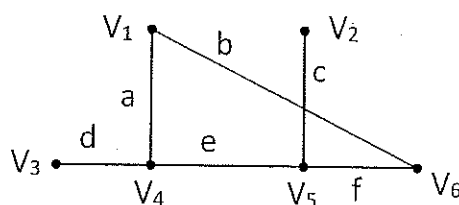
- (1) True or False: If a graph contains a sub graph homeomorphic to $K_{3,3}$ or K_5 then it is a planar graph.

- (2) True or False: A graph G is Complete if each vertex is connected to every other vertex.
- (3) True or False: A spanning tree T of graph contains all the Vertices of G .
- (4) True or False: There are 6 ways to arrange three persons in a row.
- (5) If $r < n$, then $P(n, r) = \underline{\hspace{2cm}}$
- (6) $\underline{\hspace{2cm}}$ is the number of distinct permutations that can be formed from all the letters of word CORONA.
- (7) $\underline{\hspace{2cm}}$ is the median of the observation 11, 2, 10, 4, 15, 6, 7, 18, 9.
- (8) The number of edges in the Complete graph $K_7 = \underline{\hspace{2cm}}$.

Q:3 Answer the following in short (**Attempt any Ten**).

[20]

- (1) Define complete graph. Is K_5 Complete?
- (2) Find the number n of distinct permutations that can be formed from all the letters of each word INDIAN.
- (3) A class contains 10 students with 6 men and 4 women. Find the number n of ways to select a 4-member committee with 2 men and 2 women.
- (4) Explain the positive correlation with two examples.
- (5) Mr. X buys two cows, three pigs and four hens from a man who has five cows, six pigs and eight hens. How many choices he have?
- (6) Define region of a map and degree of a region by giving example of each.
- (7) What do you mean by the chromatic number of the graph? Find the chromatic number of K_{20} .
- (8) Find cut-vertices and bridges of a graph :

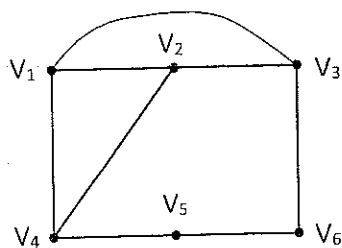


- (9) State Euler's formula with at least two examples which verifies it.
- (10) Draw a connected map of a graph with 4 vertices and 5 regions.
- (11) Find standard deviation of the observation 1, 2, 3, 4, 5, 6, 7, 8
- (12) Draw 3-regular graph with 6-vertices.

Q:4 Attempt Any Four.

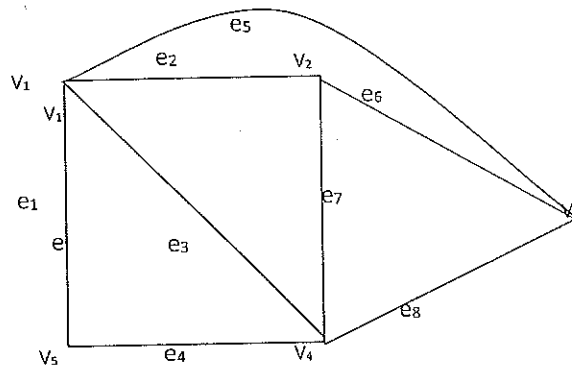
[32]

- (a) Consider the graph G as

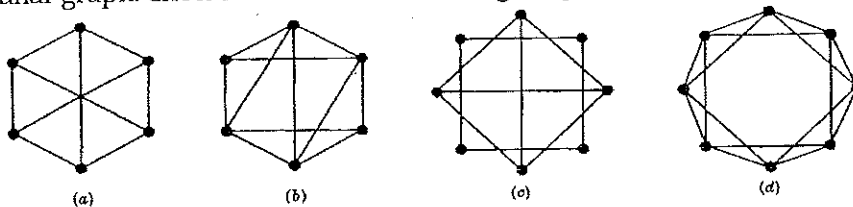


- (i) Find all simple paths from v_1 to v_6 .
- (ii) Find all trails from v_1 to v_6 .
- (iii) Find $d(v_1, v_5)$.
- (iv) Find all cycles in G .

- (b) Find the incidence matrix and adjacency matrix for the following Graphs:



- (c) Define: Planar graph. Check which of the following are planar graphs. Justify.



- (d) Describe Welch-Powell algorithm for painting a graph and give one example of it.
- (e) Suppose repetitions are not permitted. (a) Find the number of three-digit numbers that can be formed from the six digits 2, 3, 5, 6, 7, and 9. (b) How many of them are less than 400? (c) How many of them are even?
- (f) A debating team consists of 3 boys and 3 girls. Find the number of ways they can sit in a row where : (a) there are no restrictions; (b) the boys and girls are each to sit together; (c) just the girls are to sit together.
- (g) Find the standard deviation for the following frequency distribution and obtain the coefficient of variance.

x:	6	7	8	9	10	11	12
f:	3	6	9	13	8	5	4

- (h) Calculate the correlation coefficient for the following heights (in inches) of fathers (X) and their sons (Y):

X:	65	66	67	67	68	69	70	72
Y:	67	68	65	68	72	72	69	71

————— x —————

